# LINEAR ALGEBRA USING MATLAB

B.Sc. Semester V | Questions and Programs

## 1. Program on verifying basic vector space properties

```
A = [1 2 3];
B = [4 5 6];
C = [7 8 9];
k = 2;
disp('Closure (A+B):');
disp(A+B);
disp('Commutative:');
disp(isequal(A+B, B+A));
disp('Associative:');
disp(isequal((A+B)+C, A+(B+C)));
disp('Distributive:');
disp(isequal(k*(A+B), k*A + k*B));
disp('Scalar Identity:');
disp(isequal(1*A, A));
```

## 2.program on subspace

```
u = [1 -1 0];
v = [2 -2 0];
c = 2;
if sum(u+v)==0&&sum(c*u)==0
    disp('It is a SUBSPACE');
else
    disp('Not a subspace');
end
```

## 3. PROGRAM ON BASIS

```
A = [1 0 0;
0 1 0;
0 0 1];
disp('Given Matrix (Vectors as Columns):');
disp(A);
r = rank(A);
n = size(A,2);
if r == n
    disp('The given vectors are linearly independent.');
    disp('Hence, they form a basis of the vector space.');
else
    disp('The given vectors are linearly dependent.');
    disp('Hence, they do NOT form a basis of the vector space.');
end
```

## 4. Programs on Null space

```
A = [1 2 3;
4 5 6;
7 8 9];
disp('Matrix A = ');
disp(A);
N = null(A);
disp('Null Space of A = ');
disp(N);
```

## 5. Program on Row space

```
A = [1 2 3;
4 5 6;
7 8 9];
disp('Matrix A = ');
disp(A);
R = rref(A);
disp('Row Space (basis vectors) = ');
disp(R);
```

## 6. Program on Column Space

```
A = input('Enter matrix A (e.g., [1 2 3; 4 5 6; 7 8 9]): ');
[R, pivotCols] = rref(A);
col_space = A(:, pivotCols);
disp('Pivot Columns Indices:');
disp(pivotCols);
disp('Basis for Column Space:');
disp(col_space);
```

## 7. Program on dimension of a matrix

```
A = input('Enter matrix A (e.g., [1 2; 3 4]): ');
[rows, cols] = size(A);
r = rank(A);
fprintf('Number of rows = %d\n', rows);
fprintf('Number of columns = %d\n', cols);
fprintf('Dimension of column space (Rank) = %d\n', r);
```

## 8. Program on eigen vectors and eigen values

```
A = input('Enter square matrix A (e.g., [2 1; 1 2]): ');
[rows, cols] = size(A);
if rows ~= cols
    disp('Matrix must be square');
else
    [V, D] = eig(A);
    disp('Eigenvalues (Diagonal Matrix D):');
    disp(D);
    disp('Eigenvectors (Columns of V):');
    disp(V);
end
```

## 9. Program on eigen vectors and eigen values

```
A = input('Enter square matrix A (e.g., [0 -1; 1 0]): ');
[n, m] = size(A);
if n ~= m
    disp('Matrix must be square');
else
    eigenvalues = eig(A);
    real_part = real(eigenvalues);
    imag_part = imag(eigenvalues);
    figure;
    plot(real_part, imag_part, 'ro', 'MarkerSize', 8, 'LineWidth', 2);
    grid on;
    xlabel('Real Axis');
    ylabel('Imaginary Axis');
    title('Eigenvalues in Complex Plane');
    hold on;
    xline(0);
    yline(0);
    disp('Eigenvalues are:');
    disp(eigenvalues);
end
```

## 10. Program on verifying characteristic equation of a matrix

```
A = input('Enter square matrix A (e.g., [2 1; 1 2]): ');
[n, m] = size(A);
if n ~= m
    disp('Matrix must be square');
else
    p = poly(A);
    result = zeros(n);
    for i = 1:length(p)
        power = n - (i - 1);
        if power > 0
            result = result + p(i) * (A^power);
        else
            result = result + p(i) * eye(n);
        end
    end
    disp('Result of substituting A into its characteristic equation:');
    disp(result);
    if norm(result) < 1e-6
        disp('Cayley-Hamilton theorem verified (Result ≈Zero Matrix)');
    else
        disp('Verification failed');
    end
end
```

## 11. Program on diagonalization of a matrix

```
A = input('Enter the square matrix A: ');
[n, m] = size(A);
if n ~= m
    disp('Matrix must be square for diagonalization.');
    return;
end
[P, D] = eig(A);
disp('Matrix A:');
disp(A);
disp('Eigenvector matrix P:');
disp(P);
disp('Diagonal matrix D (Eigenvalues):');
disp(D);
A_check = P * D * inv(P);
disp('Verification (P * D * inv(P)) :');
disp(A_check);
if round(A_check,5) == round(A,5)
    disp('Matrix is diagonalizable.');
else
    disp('Matrix may not be diagonalizable.');
end
```

## 12. Program on orthogonal Vectors

```
A = input('Enter first vector (e.g., [1 2 3]): ');
B = input('Enter second vector (e.g., [4 5 6]): ');
if length(A) ~= length(B)
    disp('Vectors must be of the same dimension');
else
    dot_product = sum(A .* B);
    if dot_product == 0
        disp('The vectors are orthogonal');
    else
        disp('The vectors are NOT orthogonal');
    end
    fprintf('Dot Product = %d\n', dot_product);
end
```

## 13. Program on Orthonormal Vectors

```
A = input('Enter matrix (each row as a vector, e.g., [1 0 0; 0 1 0; 0 0 1]): ');
[n, m] = size(A);
isOrthonormal = true;
for i = 1:n
    if norm(A(i,:)) ~= 1
        isOrthonormal = false;
        break;
    end
end
for i = 1:n
    for j = i+1:n
        if dot(A(i, :), A(j, :)) ~= 0
            isOrthonormal = false;
            break;
        end
    end
end
if isOrthonormal
    disp('The given vectors are ORTHONORMAL');
else
    disp('The given vectors are NOT orthonormal');
end
```

## 14. Program on Gram-Schmidt process

```
A = input('Enter matrix (columns as vectors, e.g., [1 1; 1 -1]): ');
[n, m] = size(A);
Q = zeros(n, m);
for j = 1:m
    v = A(:, j);
    for i = 1:j-1
        proj = (dot(A(:, j), Q(:, i)) / dot(Q(:, i), Q(:, i))) * Q(:, i);
        v = v - proj;
    end
    Q(:, j) = v / norm(v);
end
disp('Orthonormal basis (columns):');
disp(Q);
```

## 15. Program on orthogonal projection of vectors

```
A = input('Enter vector A (e.g., [2 3 4]): ');
B = input('Enter vector B (e.g., [1 0 1]): ');
if length(A) ~= length(B)
    disp('Vectors must be of the same dimension');
else
    proj = (dot(A,B) / dot(B,B)) * B;
    disp('Projection of A onto B is:');
    disp(proj);
end
```

## 16. Write a matlab program on rank of matrix

```
A = input('Enter the matrix: ');
disp('The entered matrix is:');
disp(A);
R = rank(A);
disp('Rank of the matrix is:');
disp(R);
fprintf('The rank of the given matrix = %d\n', R);
```

## 17. Write a matlab program on orthogonal set

```
v1 = input('Enter first vector: ');
v2 = input('Enter second vector: ');
d = dot(v1,v2);
if d == 0
    disp('The vectors form an orthogonal set');
else
    disp('The vectors do not form an orthogonal set');
end
```

## 18. Write a matlab program on orthonormal set

```
v1 = input('Enter first vector: ');
v2 = input('Enter second vector: ');
if dot(v1,v2)==0&&norm(v1)==1&&norm(v2)==1
    disp('The vectors form an orthonormal set');
else
    disp('The vectors do not form an orthonormal set');
end
```

## 19. Write a matlab program to construct a diagonal matrix

```
n = input('Enter the order of the matrix: ');
v = input('Enter the diagonal elements as a vector: ');
D = diag(v);
disp('Input vector is:');
disp(v);
disp('Constructed diagonal matrix is:');
disp(D);
fprintf('Order of the diagonal matrix is %d x %d\n', n, n);
if isequal(D, diag(diag(D)))
    disp('Verification Successful: It is a diagonal matrix.');
end
```

## 20. Write a matlab program on basis

```
A = input('Enter the matrix A: ');
R = rref(A);
% Reduced Row Echelon Form
[pivotRows, pivotCols] = find(R==1);
pivotCols = unique(pivotCols);
Basis = A(:,pivotCols);
% Basis vectors
disp('The given matrix is:');
disp(A);
disp('Basis of the column space is:');
disp(Basis);
```

## 21. Write a matlab program on change of basis

```
B = input('Enter the basis matrix B: ');
v = input('Enter the vector v: ');
C = inv(B) * v;
% Coordinates in the new basis
disp('Basis matrix B = ');
disp(B);
disp('Vector coordinates in the new basis are:');
disp(C);
fprintf('Change of basis completed.\n');
```

## 22. Write a matlab program on linearly independent set

```
A = input('Enter the matrix whose columns are the vectors: ');
r = rank(A);
n = size(A,2);
if r == n
    disp('The set of vectors is linearly independent');
else
    disp('The set of vectors is linearly dependent');
end
```

## 23. Write a matlab program on linearly dependent set

```
A = input('Enter the matrix whose columns are the vectors: ');
r = rank(A);
n = size(A,2);
if r<n
    disp('The set of vectors is linearly dependent');
else
    disp('The set of vectors is linearly independent');
end
```

## 24. Write a matlab program on linear transformation

```
T = input('Enter the transformation matrix T: ');
v = input('Enter the input vector v: ');
Tv = T * v; % Linear transformation
disp('Transformation matrix T = ');
disp(T);
disp('Input vector v = ');
disp(v);
disp('Transformed vector Tv = ');
disp(Tv);
```

## 25. Write a matlab program on inner product space

```
u = input('Enter first vector u: ');
v = input('Enter second vector v: ');
ip = dot(u,v);
% Inner product
disp('First vector u = ');
disp(u);
disp('Second vector v = ');
disp(v);
fprintf('Inner product<u,v>= %f\n', ip);
if ip == 0
    disp('Vectors are orthogonal in the inner product space');
end
```